

Giacomo Medda

Resume

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Giacomo Medda
Cagliari, Italy

Research Scientist in machine learning and recommender systems, with a focus on fairness, explainability, and graph-based architectures. Experienced in developing and studying end-to-end systems involving data modeling, graph neural networks, and generative methods for explainable recommendation. Enjoys engaging in intricate challenges to find unconventional solutions and values environments where ideas are respected and collective growth is encouraged.

RESEARCH EXPERIENCE

JAN 2025 — PRESENT (FT)
University of Cagliari (Computer Science Department)
Research Collaborator

Exploring recommender system topics involving **semantic IDs**, **foundational models**, and **graph learning**. Collaborating on explainability for **LLM-profiled food** and autism-friendly POI recommendation. Coordinating student research toward publishable outcomes.

JAN 2024 — JAN 2025 (FT)
University of Cagliari (Computer Science Department)
Postdoctoral Researcher

Built an explainable food recommendation framework within the *PHASe* project, integrating **knowledge-graph reasoning** and **LLM-based data augmentation** for sustainable food choices. Deployed a modular, containerized system powering an LLM-driven conversational interface.

OCT 2020 — FEB 2024 (FT)
University of Cagliari (Computer Science Department)
Ph.D. Student

Developed **fair** and **explainable recommender** and **speaker recognition** models using **graph-based** and counterfactual methods, achieving state-of-the-art bias mitigation. Designed evaluation frameworks advancing explainability and fairness in user-centric systems.

APRIL 2022 — JULY 2022 (FT)
Eurecat (Data Science and Big Data Analytics unit)
Research Intern

Collaborated on **explainable unfairness** analysis in **GNN-based recommenders**, leading the implementation of unfairness diagnostic tools.

EDUCATION

2020 — 2024 **Doctor of Philosophy**
Mathematics and Computer Science
University of Cagliari

2018 — 2020 **Master's Degree** (MAGNA CUM LAUDE)
Computer Science
University of Cagliari

2015 — 2018 **Bachelor's Degree** (MAGNA CUM LAUDE)
Computer Science
University of Cagliari

TECHNICAL SKILLS

PyTorch •• **TensorFlow** •• **PyG** •• **scikit-learn** ••
Development and evaluation of end to end ML systems: GNNs, recommender systems, and experimental research.
Polars/Pandas •• **SciPy** •• **Matplotlib** ••
Data manipulation, EDA, statistical evaluation, and data visualization for research and prototyping.
GitHub Actions •• **FastAPI** •• **Docker** ••
Design and deployment of lightweight APIs for ML services, with automated workflows for testing and serving.
HF Transformers •• **Ollama** •• **llama-cpp** ••
Experimentation with LLM-based systems, including integration in explainable RecSys workflows, and prompt engineering for information extraction and data labeling.

APPLIED PROJECTS

hopwise — Library for explainable recommendation based on path reasoning through knowledge graphs.
PHaSE API — FastAPI service enabling zero-shot sustainable food recommendation from user preference cues.
GreenFoodLens — Dataset providing hybrid LLM-human sustainability annotations for food recommendation.

PROFESSIONAL SERVICE

Conference Organizer
UMAP 2024 — 2025, HT 2025

Program Committee & Reviewer
RecSys, ECML-PKDD, CIKM, ECIR, WSDM, WWW

Mentorship & Outreach
Thesis supervision; AI/STEM lectures (2022 — 2025)

SELECTED PUBLICATIONS

Note: In papers with alphabetical author order, I am the primary author.

[1] Balloccu, G., Boratto, L., Fenu, G., Marras, M., **Medda, G.**, Murgia, G., “GreenFoodLens: Sustainability Labels for Food Recommendation”. In: *Proceedings of the Nineteenth ACM Conference on Recommender Systems, RecSys 2025, Prague, Czech Republic, September 22-26, 2025*.

[2] **Medda, G.**, Fabbri, F., Marras, M., Boratto, L., Fenu, G., “GNNUERS: Fairness Explanation in GNNs for Recommendation via Counterfactual Reasoning”. In: *ACM Trans. Intell. Syst. Technol.* 16.1 (2025), 6:1–6:26.

[3] Boratto, L., Fabbri, F., Fenu, G., Marras, M., **Medda, G.**, “Fair Augmentation for Graph Collaborative Filtering”. In: *Proceedings of the 18th ACM Conference on Recommender Systems, RecSys 2024*. ACM.

[4] Boratto, L., Fabbri, F., Fenu, G., Marras, M., **Medda, G.**, “Counterfactual Graph Augmentation for Consumer Unfairness Mitigation in Recommender Systems”. In: *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management, CIKM 2023*. ACM.